EXAMPLES OF LOGISTIC REGRESSION

EXAMPLE-1

(a) Why is the sigmoid function used in logistic regression instead of other functions?

(b) Show that the derivative of the sigmoid function $\sigma(z) = \dfrac{1}{1+e^{-z}}$ is

$$\frac{d\sigma(z)}{dz} = \sigma(z)(1-\sigma(z)).$$

(c) Prove that the inverse of the sigmoid function is the logit function, i.e.,

$$\sigma^{-1}(p) = \log\left(\frac{p}{1-p}\right).$$

(d) A logistic regression model is used to predict whether an email is spam ($y = 1$) or not spam ($y = 0$), based on two features: keyword frequency score (x_1) and sender reputation score (x_2). Suppose we are given the following training data, where each row corresponds to an email with its x_1 and x_2 values and label y:

Email	x_1	x_2	y
1	1	17	0
2	2	25	0
3	3	40	1
4	4	35	1

The model is defined as:

$$P(y = 1 \mid x_1, x_2) = \frac{1}{1+e^{-(-2+0.5x_1+0.1x_2)}}$$

with parameters $\beta_0 = -2$, $\beta_1 = 0.5$, and $\beta_2 = 0.1$.

(i) If the logistic regression model predicts a probability of 0.87 that an email is spam, and the classification threshold is 0.5, how should the email be classified? (2)

(ii) Calculate the probability that an email with keyword frequency score $x_1 = 6$ and sender reputation score $x_2 = 40$ is spam. (2)

(iii) If the model outputs that the odds of an email being spam are $7 : 3$. What is the corresponding probability? (2)

(iv) Identify which email in the dataset the model is most uncertain about (i.e. has predicted probability closest to 0.5). (6)

(v) Suppose we change the sender reputation score for Email 2 from 25 to 40. How does the predicted probability change? Would the classification result be different? (2)

(vi) Using the logistic regression model and the given data, compute the log-likelihood function $\ell(\beta_0, \beta_1, \beta_2)$ for the given dataset. (4)

Solution

1. The sigmoid function is used in logistic regression because it maps any real number into the interval $(0, 1)$, making it a valid probability function. It is smooth, monotonic, and has a simple derivative, which simplifies optimization.

2. For $\sigma(z) = \dfrac{1}{1+e^{-z}}$:

$$\frac{d\sigma}{dz} = \frac{d}{dz}\left(1+e^{-z}\right)^{-1} = -\left(1+e^{-z}\right)^{-2} \cdot (-e^{-z}) = \frac{e^{-z}}{(1+e^{-z})^2}.$$

Since

$$\sigma(z) = \frac{1}{1+e^{-z}}, \quad 1-\sigma(z) = \frac{e^{-z}}{1+e^{-z}},$$

we obtain

$$\frac{d\sigma}{dz} = \sigma(z)(1-\sigma(z)).$$

3. Let $p = \sigma(z) = \dfrac{1}{1+e^{-z}}$. Then

$$\frac{1}{p} = 1+e^{-z}, \quad \frac{1-p}{p} = e^{-z}.$$

Taking logs:

$$z = \ln\left(\frac{p}{1-p}\right).$$

Hence, $\sigma^{-1}(p) = \log\left(\frac{p}{1-p}\right)$ (the logit function).

4. Logistic regression model:

$$P(y=1 \mid x_1, x_2) = \sigma(z), \quad z = -2 + 0.5x_1 + 0.1x_2.$$

(i) If $P = 0.87 > 0.5$, the email is classified as **spam** $(y = 1)$.

(ii) For $x_1 = 6$, $x_2 = 40$:

$$z = -2 + 0.5(6) + 0.1(40) = -2 + 3 + 4 = 5, \quad P = \sigma(5) = \frac{1}{1+e^{-5}} \approx 0.9933.$$

(iii) Odds $= 7:3 \Rightarrow \dfrac{7}{3}$. Probability:

$$p = \frac{7/3}{1+7/3} = \frac{7}{10} = 0.7.$$

(iv) Compute probabilities for dataset:

Email	x_1	x_2	y	z	$p = \sigma(z)$
1	1	17	0	0.2	0.5498
2	2	25	0	1.5	0.8176
3	3	40	1	3.5	0.9707
4	4	35	1	3.5	0.9707

Closest to 0.5 is Email 1 ($p \approx 0.5498$). Hence, the model is most uncertain about **Email 1**.

(v) For Email 2 with $x_2 = 40$:

$$z = -2 + 0.5(2) + 0.1(40) = 3, \quad p = \sigma(3) = \frac{1}{1+e^{-3}} \approx 0.9526.$$

Previously $p = 0.8176$. The probability increases, but both exceed 0.5, so the classification remains **spam**.

(vi) The log-likelihood:

$$\ell(\beta_0, \beta_1, \beta_2) = \sum_{i=1}^{n} \Big[y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i) \Big].$$

Substituting probabilities:

$$\ell = \ln(1 - 0.5498) + \ln(1 - 0.8176) + \ln(0.9707) + \ln(0.9707).$$

Numerically:

$$\ell \approx -0.7981 - 1.7007 - 0.0295 - 0.0295 = -2.5590.$$

EXAMPLE-2

A researcher used the data set to study that a randomly selected person is a smokers or non-smokers, he collected a random sample of 1,196 US males. The variables used are as follows

The variables used in the analysis are as follows:

Smoker = 1 for smokers and 0 for nonsmokers

Age = age in years

Education = number of years of schooling

Income = family income

Pcigs = price of cigarettes in individual states in 1979

The results of ML estimation for the smoker example using logit model are given in the table below,

Table 8.3 Logit model of to smoke or not to smoke.

Dependent Variable: SMOKER
Method: ML – Binary Logit (Quadratic hill climbing)
Sample: 1 1196
Included observations: 1196
Convergence achieved after 3 iterations
QML (Huber/White) standard errors & covariance

	Coefficient	Std. Error	z-Statistic	Prob.
C	2.745077	0.821765	3.340462	0.0008
AGE	−0.020853	0.003613	−5.772382	0.0000
EDUC	−0.090973	0.020548	−4.427431	0.0000
INCOME	4.72E−06	7.27E−06	0.649033	0.5163
PCIGS79	−0.022319	0.012388	−1.801626	0.0716

McFadden R-squared	0.029748	Mean dependent var	0.380435
S.D. dependent var	0.485697	S.E. of regression	0.477407
Akaike info criterion	1.297393	Sum squared resid	271.4495
Schwarz criterion	1.318658	Log likelihood	−770.8409
LR statistic	47.26785	Restr. log likelihood	−794.4748
Prob(LR statistic)	0.000000	Avg. log likelihood	−0.644516
Obs with Dep=0	741	Total obs	1196
Obs with Dep=1	455		

Let us examine these results. The variables age and education are highly statistically significant and have the expected signs. As age increases, the value of the logit decreases, perhaps due to health concerns – that is, as people age, they are less likely to smoke. Likewise, more educated people are less likely to smoke, perhaps due to the ill effects of smoking. The price of cigarettes has the expected negative sign and is significant at about the 7% level. *Ceteris paribus*, the higher the price of cigarettes, the lower is the probability of smoking. Income has no statistically visible impact on smoking, perhaps because expenditure on cigarettes may be a small proportion of family income.

The interpretation of the various coefficients is as follows: holding other variables constant, if, for example, education increases by one year, the average logit value goes down by ≈ 0.09, that is, the log of odds in favor of smoking goes down by about 0.09. Other coefficients are interpreted similarly.

But the logit language is not everyday language. What we would like to know is the probability of smoking, given values of the explanatory variables. But this can be computed from Eq. (8.7). To illustrate, take smoker #2 from **Table 8.1** also. His data are as follows: age = 28, educ = 15, income = 12,500 and pcigs79 = 60.0. Inserting these values in Eq. (8.7), we obtain:

$$P = \frac{1}{1 + e^{-(-0.4935)}} \approx 0.3782$$

That is, the probability that a person with the given characteristics is a smoker is about 38%. From our data we know that this person is a smoker.

Now take a person with age, educ, income, and pcigs79 of 63, 10, 20,000, and 60.8, respectively. For this person, the probability of smoking is

$$P = \frac{1}{1 + e^{-(-0.7362)}} = 0.3227$$

That is, the probability of this person being a smoker is 32%. In our sample such a person is nonsmoker.